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Lab Task No 1

```python
#tASK 1
odd_squares = [x**2 for x in range(10) if x %2==1]
print(odd_squares)
odd_numbers = [x for x in range(10) if x % 2 == 0]
print(odd_numbers)
unique_list=list(set(odd_squares))
print(unique_list)
final_list=sorted(unique_list, reverse = True)
print(final_list)
```

```
[1, 9, 25, 49, 81]
[0, 2, 4, 6, 8]
[1, 9, 81, 49, 25]
[81, 49, 25, 9, 1]
```

Lab Task No 2

```python
#tASK 2
def generate_list():
  squares=[x **2 for x in range(1,31)]
  return squares
my_list = generate_list()
print(my_list[:7])
print(my_list[-7:])
print(my_list[7:])
even_list=[x for x in my_list if x % 2 ==0]
print(even_list)
```

```
[1, 4, 9, 16, 25, 36, 49]
[576, 625, 676, 729, 784, 841, 900]
[64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400, 441, 484, 529, 576, 625,
676, 729, 784, 841, 900]
[4, 16, 36, 64, 100, 144, 196, 256, 324, 400, 484, 576, 676, 784, 900]
```

Lab Task No 3

```python
#tASK 3
number =input("enter number :")
data = [int(x.strip()) for x in number.split(',')]
num=[]
for x in data :
  if x == 0:
    break
  num.append(x)
reversed_list = num[::-1]
for num in reversed_list:
  print(num)
```

```
enter number :10 , 20 , 30 , 40 , 50
50
40
30
20
10
```

Lab Task No 4

```python
#tASK 4
negative =[]
zero = []
positive = []
while True :
  x= input("enter a number:")
  if x=="":
    break
  x = int(x)
  if x < 0:
    negative.append(x)
  elif x == 0:
    zero . append(x)
  else:
    positive.append(x)
for i in negative + zero +positive:
  print (i)

```

```
enter a number:0
enter a number:-2
enter a number:4
enter a number:0
enter a number:-1
enter a number:0
enter a number:-3
enter a number:2
enter a number:-1
enter a number:
-2
-1
-3
-1
0
0
0
4
2
```